

№ 1

а) Пусть $E_n = \{x \in E \mid |f(x)| \geq \frac{1}{n}\}$, $n \in \mathbb{N}$.

$$0 = \int_E |f(x)| d\mu(x) \geq \int_{E_n} |f(x)| d\mu(x) \geq \int_{E_n} \frac{1}{n} d\mu(x) = \frac{1}{n} \mu(E_n) \implies \\ \implies \frac{1}{n} \mu(E_n) \leq 0, \quad \mu(E_n) \geq 0, \quad \frac{1}{n} > 0 \implies \mu(E_n) = 0$$

$$\{x \in E \mid |f(x)| > 0\} = \bigcup_{n=1}^{\infty} E_n \leq \sum_{n=1}^{\infty} \mu(E_n) = 0 \implies f(x) = 0 \text{ п.в.}$$

б) $f = f^+ - f^-$

$A^+ = \{x \in E \mid f(x) \geq 0\}$, и из (а): $f(\chi_{E \setminus A^+}) = 0$ п.в. $\implies \mu(\{x \in E \mid f(x) > 0\}) = 0$;

$A^- = \{x \in E \mid f(x) < 0\}$

$$\int_{A^-} f(x) d\mu = 0 \implies \int_{A^-} (-f(x)) d\mu = 0 \implies f(x) = 0 \text{ п.в. на } A^-$$

№ 2

1) $A = \{0, 1\}$, $B = (0, 1)$

$$d_x(A) = 0, \quad d_{[x]}(B) = 0 \\ d_x(B) = 1, \quad d_{[x]}(A) = 1$$

2) $A = C$

$B = [0, 1] \setminus C$

$$d_x(A) = 0 \quad d_x(B) = 1 \\ d_\lambda(B) = 0 \quad d_\lambda(A) = 1$$

3) $A = \{0, 1\}$

$B = (0, 1)$

$$d_\lambda(A) = d_\lambda(\{0\}) + d_\lambda(\{1\}) = 0 \\ d_{[x]}(B) = 0 \\ d_\lambda(B) = 1 \\ d_{[x]}(A) = 1$$

№ 3

$$\mu_F((a, b]) = F(b) - F(a) = \int_a^b F'(x) dx = \int_{(a, b]} F'(x) dx$$

Теорема Леви: $(X, \mathcal{A}, \mu); f_n: X \rightarrow [0, +\infty), 0 \leq f_1 \leq f_2 \leq \dots$ п.в. $x \in X$.
 Если $f(x) = \lim_{n \rightarrow +\infty} f_n(x)$ п.в. $x \in X$, то $f(x)$ — измерима, и

$$\int_X f d\mu = \lim_{n \rightarrow +\infty} \int_X f_n d\mu$$

Вычисление интеграла:

$$\begin{aligned} \int_{\mathbb{R}} f(x) d\mu_F(x) &= \lim_{n \rightarrow +\infty} \left(\int_{\mathbb{R}} f_n(x) d\mu_F(x) \right) = \lim_{n \rightarrow +\infty} \left(\int_{\mathbb{R}} \sum_{k=1}^m c_k \mathbb{I}_{E_k}(x) d\mu(x) \right) = \\ &= \lim_{n \rightarrow +\infty} \left(\sum_{k=1}^m c_k \mu_F(E_k) \right) = \lim_{n \rightarrow +\infty} \left(\sum_{k=1}^m c_k \int_{E_k} F'(x) dx \right) = \\ &= \lim_{n \rightarrow +\infty} \left(\sum_{k=1}^m c_k \int_{\mathbb{R}} \mathbb{I}_{E_k}(x) F'(x) dx \right) = \lim_{n \rightarrow +\infty} \left(\int_{\mathbb{R}} f_n(x) F'(x) dx \right) = \int_{\mathbb{R}} f(x) F'(x) dx \end{aligned}$$

№ 4

$$F_1(x) + C = F_2(x), \quad C \in \mathbb{R}$$